

MUHAMMAD AMINU

(Mechatronics engineering student)

- Phone: 08060484117
- Email: afamsmuhd@gmail.com
- Locations: Durumi Extension, Abuja state, FCT
- LinkedIn/GitHub: MuhdAM (github.com/MuhdAM | linkedin.com/in/MuhdAM)

Career Objective / Personal Statement: Motivated 300-level Mechatronics Engineering student at Air Force Institute of Technology (AFIT) seeking a 3-month SIWES placement to apply hands-on skills in AI, robotics, and automation. Passionate about inclusive/all-in technology solutions for Nigeria, with projects in Hausa-speaking AI and embedded robotics. Eager to contribute to NCAIR's robotics and multilingual AI initiatives (like N-ATLAS) while gaining practical experience in a professional environment.

EDUCATION

- Air Force Institute of Technology (AFIT), Kaduna {B.Eng. Mechatronics Engineering}
- Relevant Coursework: Embedded Systems, Automation, CAD (AutoCAD/SolidWorks), Basic Electronics, Programming
- Current Level: 300L | CGPA: 4.6/5.0

PROJECTS

- NomaScout Autonomous Rover (personal project, 2026) - Build and programmed offline ESP32-CAM 4WD rover for smallholder farmers. Features real-time crop detection, autonomous navigation, and Hausa voice alerts. Submitted to AMTES 2026 inter-Varsity competition. Video & GitHub available upon request
- Eco-Bridge AI Chatbot (Personal/Team Project, 2025–Present) - Developed Hausa-speaking AI chatbot with SpeechT5 TTS and 3D avatar for inclusive access. Addressed digital language gap for 50M+ Hausa speakers. **GitHub:** <https://github.com/MuhdAM/EcoBridge-Chatbot>.
- StudySpark Gamified Study App (Team/Personal, Ongoing) - AI-powered tool with notes, practice questions, XP points, streaks, and Leader-boards. Focused on improving student consistency and motivation. **GitHub:** Repo available upon request

SKILLS:

- Programming: Python (Intermediate), Arduino/ESP32 Programming
- Software: AutoCAD, SolidWorks (Basics), Git
- Hardware: Basic Electronics, Sensor Integration (Ultrasonic, Camera)
- Other: Computer Vision Basics (Self-Learning), AI/ML Concepts, Teamwork, Problem-Solving

EXPERIENCE / TRAINING

- SWEP (Student Workshop Experience Practice)/Lab Sessions (AFIT, 200L): Hands-on introduction to Arduino, SolidWorks, AutoCAD, and robotic arm demonstration.
- Computer training at SafrimICT, Zamfara. (2020)

CERTIFICATIONS / EXTRACURRICULAR

- Diploma in Computer Science at SafrimICT Zamfara (2020); • Cowrywise Ambassador